



DEPARTMENT OF  
COMPUTER SCIENCE



# ABSTRACTION

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## LEARNING OBJECTIVES

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At the end of the lesson, you should be able to:

- Discuss abstraction and abstraction types.
- Apply access specifiers to computer programmes.
- Develop a simple computer programme in C.



## ABSTRACTION: AN ANALOGY

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A real world example of how abstraction works is when turn on the ignition of a car.

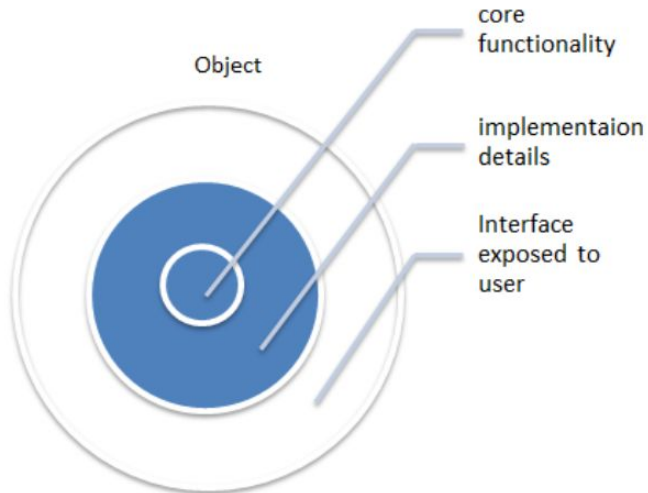
We do not bother ourselves down with what happens in the engine in between the time we twist the car key to the ignition point and when the car actually starts.



## ABSTRACTION

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Abstraction is the process of hiding the internal details of an application from the outer world.





## DATA ABSTRACTION

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When we write our programme in such a way that the data of an object is not directly exposed to the outside world or to other programmes, we have implemented data abstraction.

Doing this also ensures that no other programme can alter the object data without obtaining the necessary permissions.



## FUNCTIONAL ABSTRACTION

It involves writing a programme in such a way that the implementation of the methods of an object is not directly exposed to the outside world or to other programmes.

If other programmes need to use the method of an object, all they need to know is the method's name, the format of the input, and the format of the output.



## **IMPLEMENTING ABSTRACTION: ACCESS SPECIFIERS**

We can implement abstraction by using access specifiers such as public and private for the properties and functions of an object.

The public and private access specifiers restrict the access of other programmes to only some of the properties and functions of an object



## IMPLEMENTING ABSTRACTION: ACCESS SPECIFIERS

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```
class Car
{
    // Variables
    private char make[20];
    private char colour[10];
    private char registration_number[10];

    // Method
    public char[] get_make() {
        // Get car make here
    }
}
```

```
class Car
{
    // Method
    private void create_spark() {
        // Create spark here;
    }

    // Method
    public void turn_on(char[] _name) {
        // Turn car engine on here;
    }
}
```



## IMPLEMENTING ABSTRACTION: HEADER FILES

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We can also implement abstraction with header files.

By including a header file as part of our code, we can easily use functions from the header files without having any knowledge of how the function is implemented in the header file.

```
#include <stdio.h>

int main( ) {

    printf( "I am a computer program");

    return 0;

}
```



## BENEFITS OF ABSTRACTION

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- It ensures that our applications are more secure since the implementation of the process is not exposed to the outside world.
- It enhances code readability and reusability.



## **BENEFITS OF ABSTRACTION**

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- It ensures that we can change the implementation of a method without breaking other parts of the code.
- It prevent the duplication of codes.



## A HELLO WORLD PROGRAM

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- Comments in C start with /\* and are terminated with \*/
- Header files contain the information necessary to use these libraries, such as function declarations.

```
/* A Hello World Program */  
#include<stdio.h>  
  
int main() {  
    printf("Hello World!\n");  
}
```



## A HELLO WORLD PROGRAM

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- All C programmes have `main()` as the entry-point function.
- The braces `{` and `}` describes the start and end of the function block.

```
/* A Hello World Program */  
#include<stdio.h>  
  
int main() {  
    printf("Hello World!\n");  
}
```



## A HELLO WORLD PROGRAM

- This programme contains just one statement: a function call to the standard library function `printf()`, which prints a character string to standard output (usually the screen).

```
/* A Hello World Program */  
#include<stdio.h>  
  
int main() {  
    printf("Hello World!\n");  
}
```



## ESCAPE SEQUENCES AND MEANING

| Escape Sequences | Meaning         |
|------------------|-----------------|
| \'               | Single Quote    |
| \"               | Double Quote    |
| \\               | Backslash       |
| \0               | Null            |
| \a               | Bell            |
| \b               | Backspace       |
| \f               | form Feed       |
| \n               | Newline         |
| \r               | Carriage Return |
| \t               | Horizontal Tab  |
| \v               | Vertical Tab    |



## SUMMARY

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- Abstraction is the process of hiding the internal details of an application from the outer world.
- All C programmes have `main()` as the entry-point function.
- Abstraction ensures that we can change the implementation of a method without breaking other parts of the code.



## FURTHER READING RESOURCES

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- Bailey, T. (2005). An introduction to the c programming language and software design. July-2005
- Kernighan, B. W., & Ritchie, D. M. (2002). The C programming language.



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THANK  
YOU